Project Proposal: AI Image Quality Enhancement Pipeline

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Section 1: The Problem Definition (Domain & Theme)

I have selected **Theme A: Image Processing.** We will implement a pipeline consisting of: [(1) contrast adjustment, (2) gaussian blur, and (3) image upscaling]. These programs will operate sequentially, with each of their outputs feeding into the next.

Section 2: The Data Strategy (Input & Output)

I will use the *RealVisXL V5.0* AI image generator as our source, found on CivitAI at https://civitai.com/models/139562/realvisxl-v50. The model will run V5.0 Lightning (BakedVAE), and will not run on any LoRa's nor any post-processing adjustments other than what has been hand-developed for this project.

The images produced from RealVisXL V5.0 will be designated in order as small, medium, and large: [(1024 x 1024) images, (1024 x 1792) images, (1334 x 2368) images]. These numbers represent pixel counts and were chosen to work effectively with the AI image generation model. Small inputs are for debugging. Large inputs are to stress-test the MPU communication overhead.

These programs will be written in C and used in a ComfyUI pipeline as secondary nodes in the pipeline. The C files will be submitted as work for the final project, as opposed to the ComfyUI node code. However, in the interest of having an interesting presentation, I aim to have a live demonstration (with help of the audience suggesting prompts) to create a safe-for-work AI image, to be passed through the image enhancement pipeline in real-time.

Section 3: Goals & Verification Plan

Success Criteria: My program will output a beautiful .jpg file image.

Goals: Implement Serial, OpenMP, and MPI versions.

Verification: I will write a script to compare the binary output of the parallel versions against the serial baseline to ensure they are identical.

Performance: I aim to achieve a near-linear speedup on OpenMP but expect diminishing returns on MPI due to network latency.